INSTITUTO FEDERAL
São Paulo
Câmpus Hortolândia

Engenharia de Controle e Automação

MIC - Microcontroladores

Arquitetura dos microcontroladores AVR

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Rev.2026.1

Estudando o datasheet do ATMega328

MICROCHIP

ATmega48A/PA/88A/PA/168A/PA/328/P

megaAVR® Data Sheet

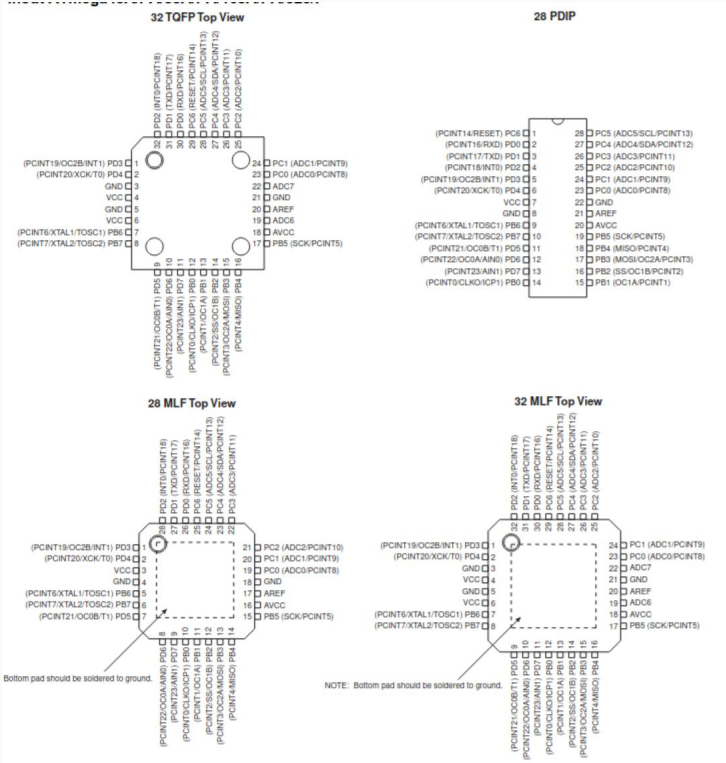
Introduction

The ATmega48A/PA/88A/PA/168A/PA/328/P is a low power, CMOS 8-bit microcontrollers based on the AVR® enhanced RISC architecture. By executing instructions in a single clock cycle, the devices achieve CPU throughput approaching one million instructions per second (MIPS) per megahertz, allowing the system designer to optimize power consumption versus processing speed.

- Família de microcontroladores
- Baixo consumo (low power)
- 8 bits CMOS
- Baseados na arquitetura RISC melhorado (enhanced RISC)
- Executa instruções em um único ciclo de clock
- Otimiza consumo de energia e velocidade de processamento

Estudando o datasheet do ATMega328

Encapsulamentos



TQFP (SMD) 32 pinos

PDIP - 28 pinos

Utilizado no Arduino UNO

MLF 28 pinos

MLF (SMD) 32 pinos

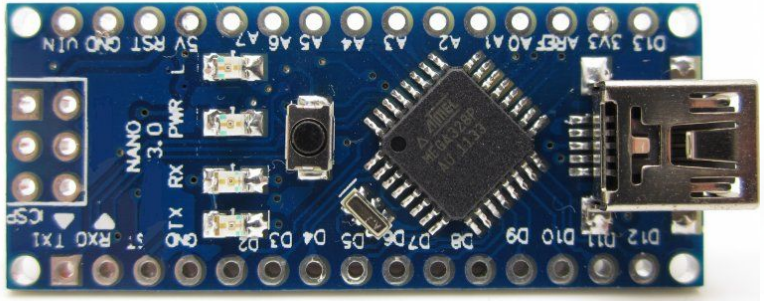
TQFP e MLF de 32 pinos tem 2 conversores AD a mais que os de 28 pinos

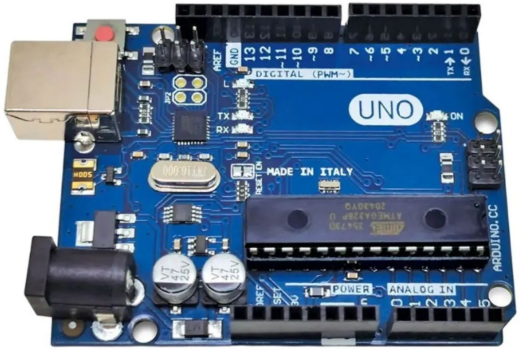
Aplicações - exemplos

TQFP (SMD) 32 pinos

Utilizado no Arduino Nano e UNO modelo mais novo

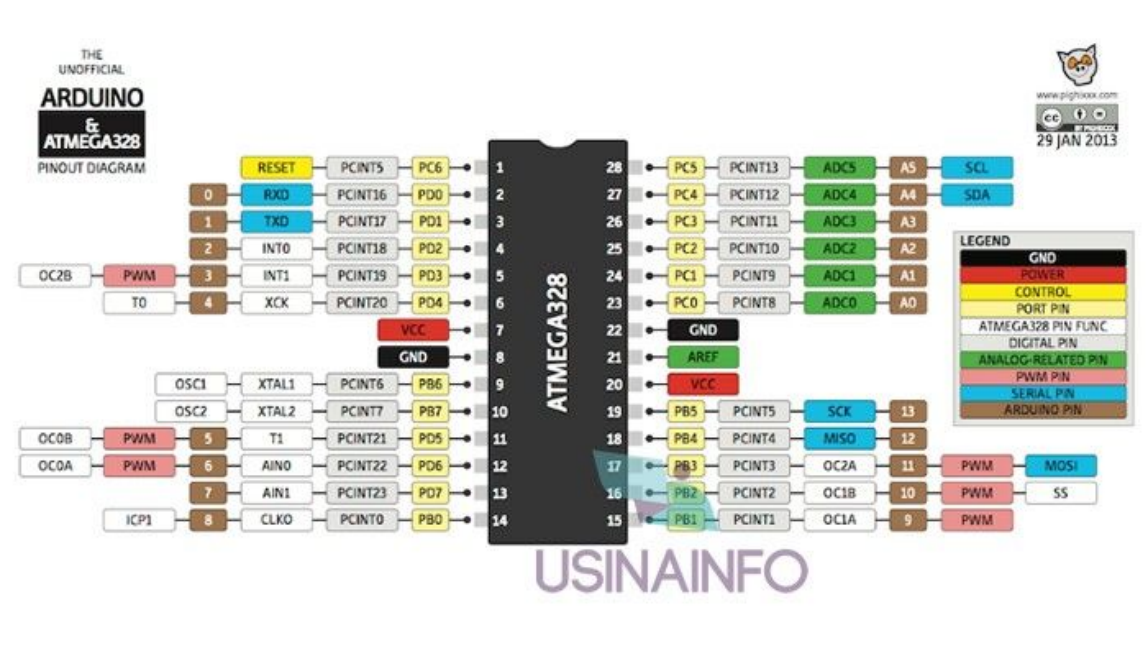
PDIP - 28 pinos (Fora de linha)





Utilizado no Arduino UNO modelo antigo (prefira este!)

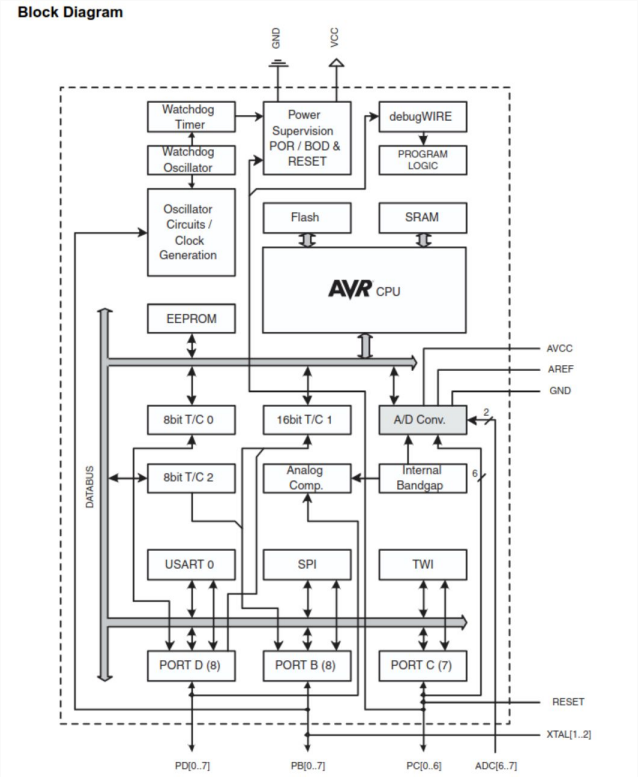
Pinagem ATMega328 PDIP



Atenção

- Pinos VCC, GND, AREF tem **função única**
- Demais pinos tem várias funções (multiplexados)
- Definição de qual função do pino definida no projeto do sistema embarcado

Diagrama em blocos - Datasheet



Observar:

- Bloco CPU AVR - Executa o processamento
- Parte pontilhada no diagrama é o microcontrolador - CPU + periféricos
- Pinos externos - AVCC, AREF, GND, VCC, GND, RESET, XTAL (cristal oscilador)
- Observe as portas de acesso aos periféricos - multiplexados: PortB[0..7], PortC[0..6], PortD[0..7] e ao ADC[6,7]
- Barramento único de dados - **databus de 8 bits** - Define a família como sendo de 8 bits
- Todos os dados circulam por este barramento - interliga todos os periféricos

Comparação entre os microcontroladores da família

Comparison Between Processors

The ATmega48A/PA/88A/PA/168A/PA/328/P differ only in memory sizes, boot loader support, and interrupt vector sizes. Table 2-1 summarizes the different memory and interrupt vector sizes for the devices.

Table 2-1. Memory Size Summary

Device	Flash	EEPROM	RAM	Interrupt Vector Size
ATmega48A	4KBytes	256Bytes	512Bytes	1 instruction word/vector
ATmega48PA	4KBytes	256Bytes	512Bytes	1 instruction word/vector
ATmega88A	8KBytes	512Bytes	1KBytes	1 instruction word/vector
ATmega88PA	8KBytes	512Bytes	1KBytes	1 instruction word/vector
ATmega168A	16KBytes	512Bytes	1KBytes	2 instruction words/vector
ATmega168PA	16KBytes	512Bytes	1KBytes	2 instruction words/vector
ATmega328	32KBytes	1KBytes	2KBytes	2 instruction words/vector
ATmega328P	32KBytes	1KBytes	2KBytes	2 instruction words/vector

ATmega**328** - 32 KBytes de memória flash (programa) e arquitetura de 8 bits - **Databus de 8 bits**

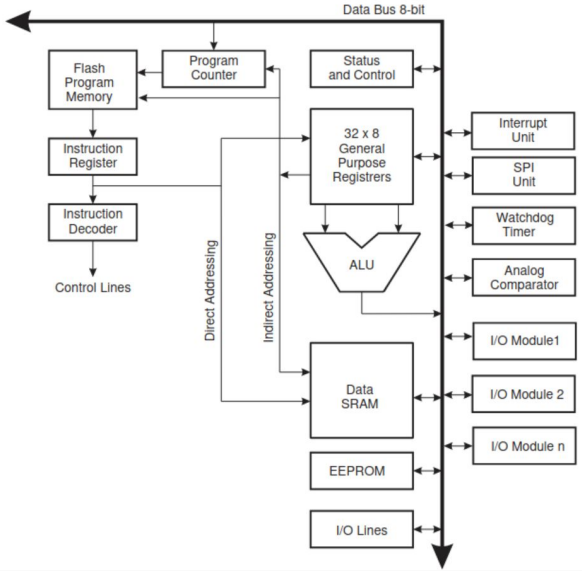
Visão geral - AVR Core (processador AVR) - cap. 7

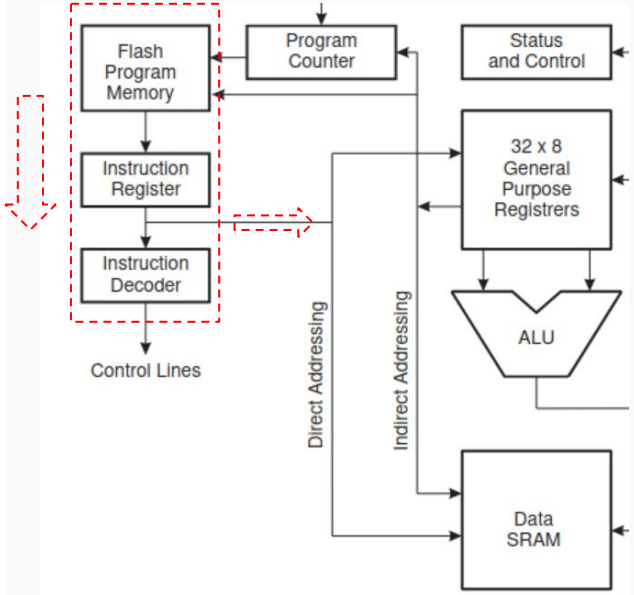
7. AVR CPU Core

7.1 Overview

This section discusses the AVR core architecture in general. The main function of the CPU core is to ensure correct program execution. The CPU must therefore be able to access memories, perform calculations, control peripherals, and handle interrupts.

Figure 7-1. Block Diagram of the AVR Architecture





32 registradores de uso geral de 8 bits

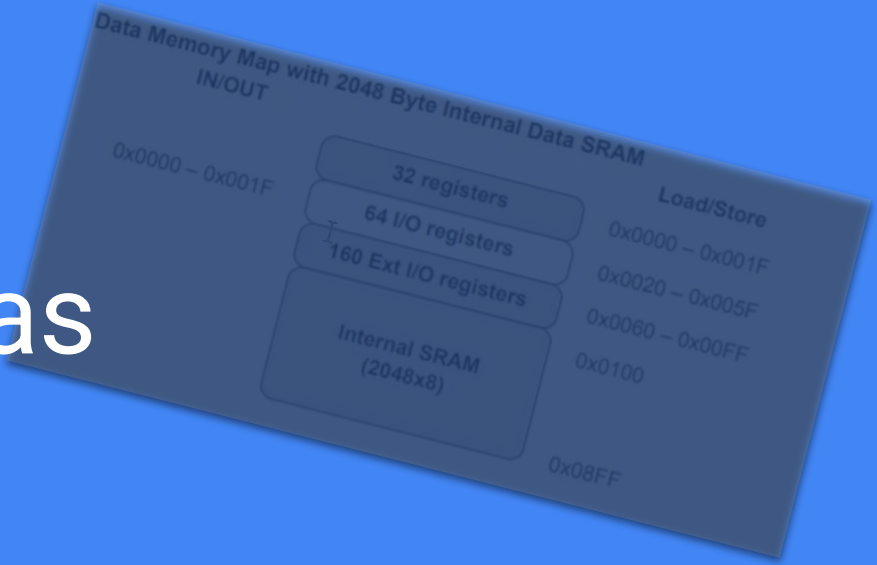
ULA - recebe dados destes registradores, somente

Instruções são lidas da memória Flash (de programa) e armazenadas no registrador de instruções

A partir daí são dois caminhos:

- Acessam a memória de dados (SRAM) ou registradores
- São decodificadas e acionam as linhas de controle **(Control lines)**

Memórias internas



Memórias do ATMega328 - Registradores de Uso Geral (GPRs)

Figure 7-2. AVR CPU General Purpose Working Registers

	7 … 0	Addr.	
General Purpose Working Registers	R0	0x00	
	R1	0x01	
	R2	0x02	
	...		
	R13	0x0D	
	R14	0x0E	
	R15	0x0F	
	R16	0x10	
	R17	0x11	
	...		
	R26	0x1A	X-register Low Byte
	R27	0x1B	X-register High Byte
	R28	0x1C	Y-register Low Byte
	R29	0x1D	Y-register High Byte
	R30	0x1E	Z-register Low Byte
	R31	0x1F	Z-register High Byte

Cada registrador:

- 8 bits
- Endereços de 0x00 a 0x1F (**32 registradores no total)**
- Maior parte dos registradores compatíveis com todas instruções de assembler (asm)
- Alguns não são compatíveis com todas instruções asm

- R26 - R27 - ponteiro X
- R28 - R29 - ponteiro Y
- R30 - R31 - ponteiro Z

Realizam operações de 16 bits para acessar diretamente a memória de dados

Sobre os ponteiros X, Y e Z

Memória de dados 2048 posições - São necessários 11 bits para acessar todas as posições, daí a necessidade de utilizar 2 pares de registradores para endereçamento

Memórias do ATMega328 - Memória de programa - flash (cap.8)

8. AVR Memories

8.1 Overview

This section describes the different memories in the ATmega48A/PA/88A/PA/168A/PA/328/P. The AVR architecture has two main memory spaces, the Data Memory and the Program Memory space. In addition, the ATmega48A/PA/88A/PA/168A/PA/328/P features an EEPROM Memory for data storage. All three memory spaces are linear and regular.

8.2 In-System Reprogrammable Flash Program Memory

The ATmega48A/PA/88A/PA/168A/PA/328/P contains 4/8/16/32Kbytes On-chip In-System Reprogrammable Flash memory for program storage. Since all AVR instructions are 16 or 32 bits wide, the Flash is organized as 2/4/8/16K x 16. For software security, the Flash Program memory space is divided into two sections, Boot Loader Section and Application Program Section in ATmega88PA and ATmega168PA. See SPMEN description in section "SPMCSR – Store Program Memory Control and Status Register" on page 287 for more details.

The Flash memory has an endurance of at least 10,000 write/erase cycles. The ATmega48A/PA/88A/PA/168A/PA/328/P Program Counter (PC) is 11/12/13/14 bits wide, thus addressing the 2/4/8/16K program memory locations. The operation of Boot Program section and associated Boot Lock bits for software protection are described in detail in "Self-Programming the Flash, ATmega 48A/48PA" on page 264 and "Boot Loader Support – Read-While-Write Self-Programming" on page 272. "Memory Programming" on page 289 contains a detailed description on Flash Programming in SPI- or Parallel Programming mode.

Constant tables can be allocated within the entire program memory address space (see the LPM – Load Program Memory instruction description).

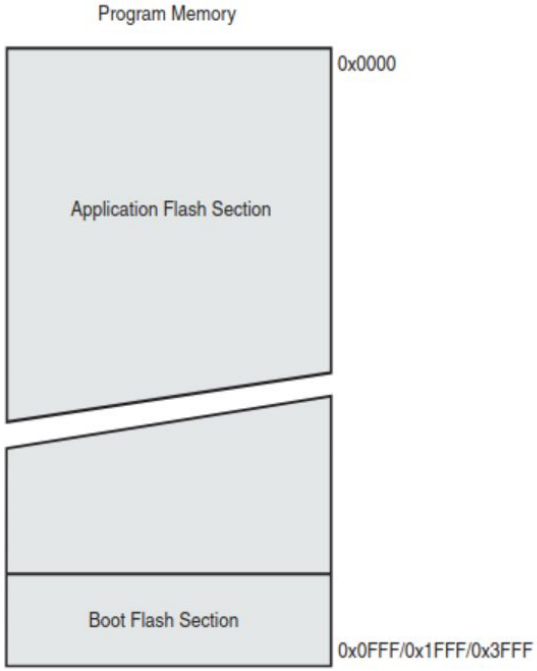
Timing diagrams for instruction fetch and execution are presented in "Instruction Execution Timing" on page 23.

8.2 - Memória de programa - Flash

- Atmega328 - 32 Kbytes
- Instruções do AVR tem 16 ou 32 bits de comprimento
- Memória flash organizada em 16K x 16 bits (2 setores de 8 bits)
- Logo, p**rogramas de até 16Kbytes são aceitos neste microcontrolador**
- **16384 endereços, cada endereço com 2 bytes**

Memórias do ATMega328 - Memória de programa - flash (cap.8)

Figure 8-2. Program Memory Map ATmega88A, ATmega88PA, ATmega168A, ATmega168PA, ATmega328 and ATmega328P



8.2 - Memória de programa - Flash - Organização - atmega328

- Endereço: 0x0000 a 0x3FFF (16k)
- Programa até 16Kbytes

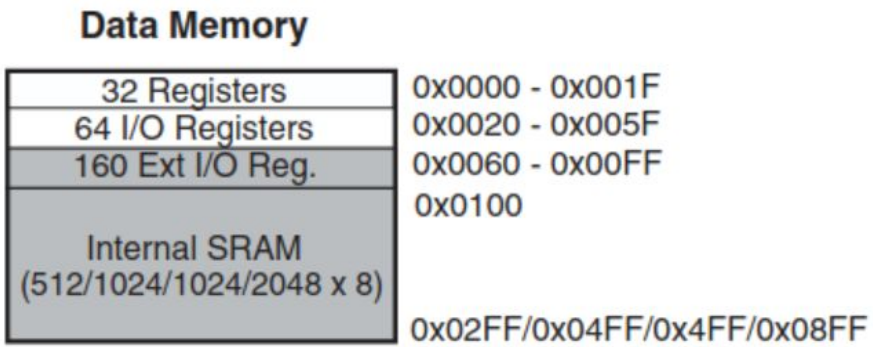
- O programa pode ter até 16K ou posso definir um espaço para um **bootloader** (boot Flash Section)

- O **bootloader** é um programa que funciona como um gravador de programa. Isso permite a gravação de código diretamente a partir de um cabo USB **(Arduino)**

8.3 - Memória de dados - SRAM - atmega328

- Perde o conteúdo quando o CI é desligado
- 2048 x 8 - posições disponíveis organizadas conforme este mapa:
- 32 Registradores de uso geral - 0x0000 a 0x001**F**
- 64 Registradores de uso específico - 0x0020 a 0x005**F**
- 160 bytes espaço reservado - 0x0060 a 0x00**FF**
- **Restante da memória - SRAM interna - variáveis do programa - 0x0100 a 0x08FF**

Figure 8-3. Data Memory Map



8.3 - Memória de dados - SRAM - atmega328

- Registradores de uso específico

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Page
0x15 (0x35)	TIFR0	–	–	–	–	–	OCF0B	OCF0A	TOV0	
0x14 (0x34)	Reserved	–	–	–	–	–	–	–	–	
0x13 (0x33)	Reserved	–	–	–	–	–	–	–	–	
0x12 (0x32)	Reserved	–	–	–	–	–	–	–	–	
0x11 (0x31)	Reserved	–	–	–	–	–	–	–	–	
0x10 (0x30)	Reserved	–	–	–	–	–	–	–	–	
0x0F (0x2F)	Reserved	–	–	–	–	–	–	–	–	
0x0E (0x2E)	Reserved	–	–	–	–	–	–	–	–	
0x0D (0x2D)	Reserved	–	–	–	–	–	–	–	–	
0x0C (0x2C)	Reserved	–	–	–	–	–	–	–	–	
0x0B (0x2B)	PORTD	PORTD7	PORTD6	PORTD5	PORTD4	PORTD3	PORTD2	PORTD1	PORTD0	101
0x0A (0x2A)	DDRD	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	101
0x09 (0x29)	PIND	PIND7	PIND6	PIND5	PIND4	PIND3	PIND2	PIND1	PIND0	101
0x08 (0x28)	PORTC	–	PORTC6	PORTC5	PORTC4	PORTC3	PORTC2	PORTC1	PORTC0	100
0x07 (0x27)	DDRC	–	DDC6	DDC5	DDC4	DDC3	DDC2	DDC1	DDC0	100
0x06 (0x26)	PINC	–	PINC6	PINC5	PINC4	PINC3	PINC2	PINC1	PINC0	101
0x05 (0x25)	PORTB	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	100
0x04 (0x24)	DDRB	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	100
0x03 (0x23)	PINB	PINB7	PINB6	PINB5	PINB4	PINB3	PINB2	PINB1	PINB0	100
0x02 (0x22)	Reserved	–	–	–	–	–	–	–	–	
0x01 (0x21)	Reserved	–	–	–	–	–	–	–	–	
0x0 (0x20)	Reserved	–	–	–	–	–	–	–	–	

Portas de E/S

Timers

(0x8C)	Reserved	– – – – – – – –	
(0x8B)	OCR1BH	Timer/Counter1 - Output Compare Register B High Byte	144
(0x8A)	OCR1BL	Timer/Counter1 - Output Compare Register B Low Byte	144
(0x89)	OCR1AH	Timer/Counter1 - Output Compare Register A High Byte	144
(0x88)	OCR1AL	Timer/Counter1 - Output Compare Register A Low Byte	144
(0x87)	ICR1H	Timer/Counter1 - Input Capture Register High Byte	144
(0x86)	ICR1L	Timer/Counter1 - Input Capture Register Low Byte	144
(0x85)	TCNT1H	Timer/Counter1 - Counter Register High Byte	143
(0x84)	TCNT1L	Timer/Counter1 - Counter Register Low Byte	143
(0x83)	Reserved	– – – – – – – –	

Sistemas de clock e reset

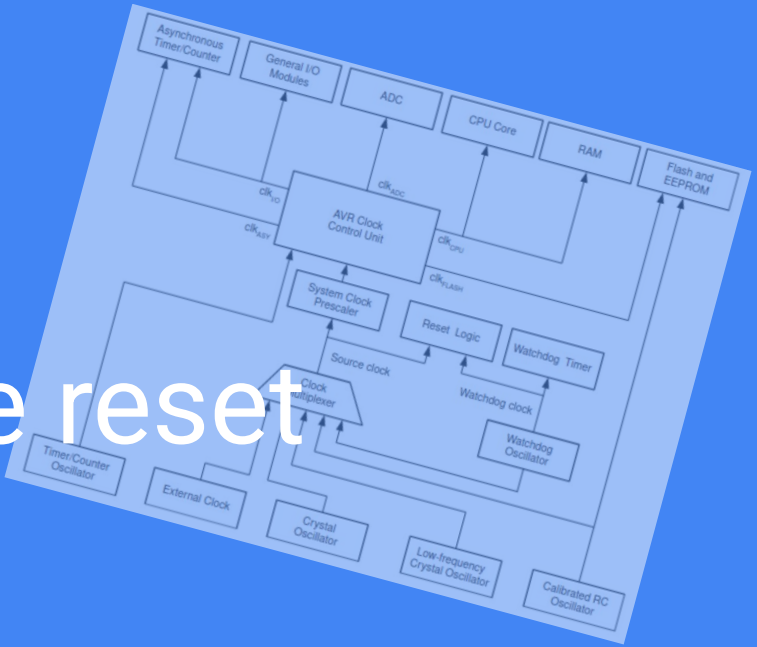


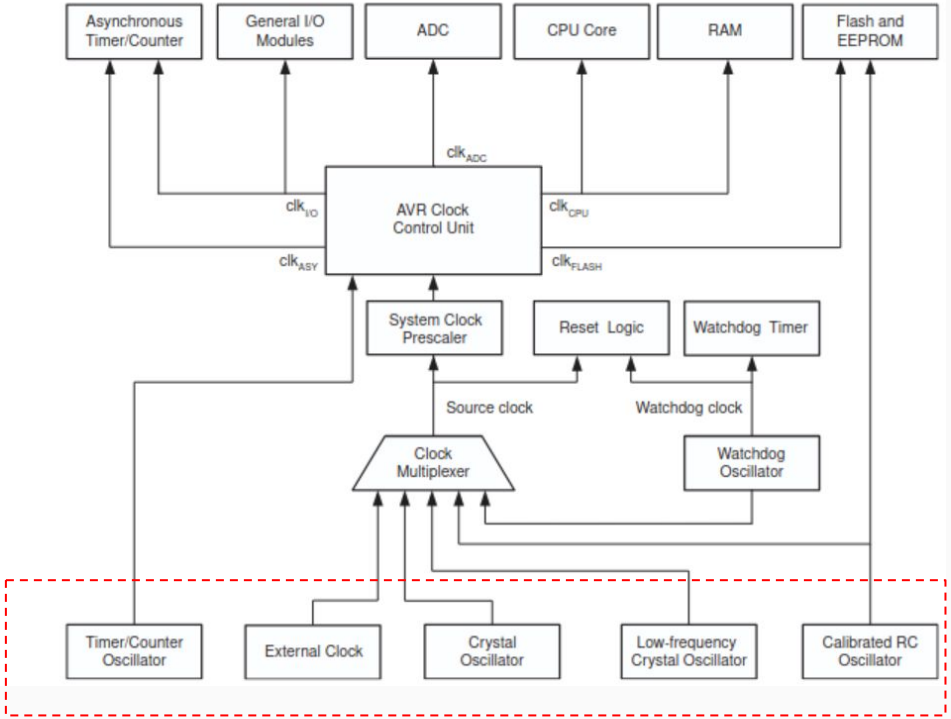
Table 9-1. Device Clocking Options Select[1]

Device Clocking Option	CKSEL3...0
Low Power Crystal Oscillator	1111 - 1000
Full Swing Crystal Oscillator	0111 - 0110
Low Frequency Crystal Oscillator	0101 - 0100
Internal 128kHz RC Oscillator	0011
Calibrated Internal RC Oscillator	0010
External Clock	0000
Reserved	0001

Note: 1. For all fuses “1” means unprogrammed while “0” means programmed.

ATMega328 - Sistema de clock

Figure 9-1. Clock Distribution



Cristal oscilador

- Frequências mais altas
- Maior precisão

Oscilador RC interno

- 8 MHz
- Circuito mais simples
- Período varia mais com a temperatura

Cristal oscilador de baixa frequência

- Frequências mais baixas
- Bastante Precisos
- Menor consumo de energia

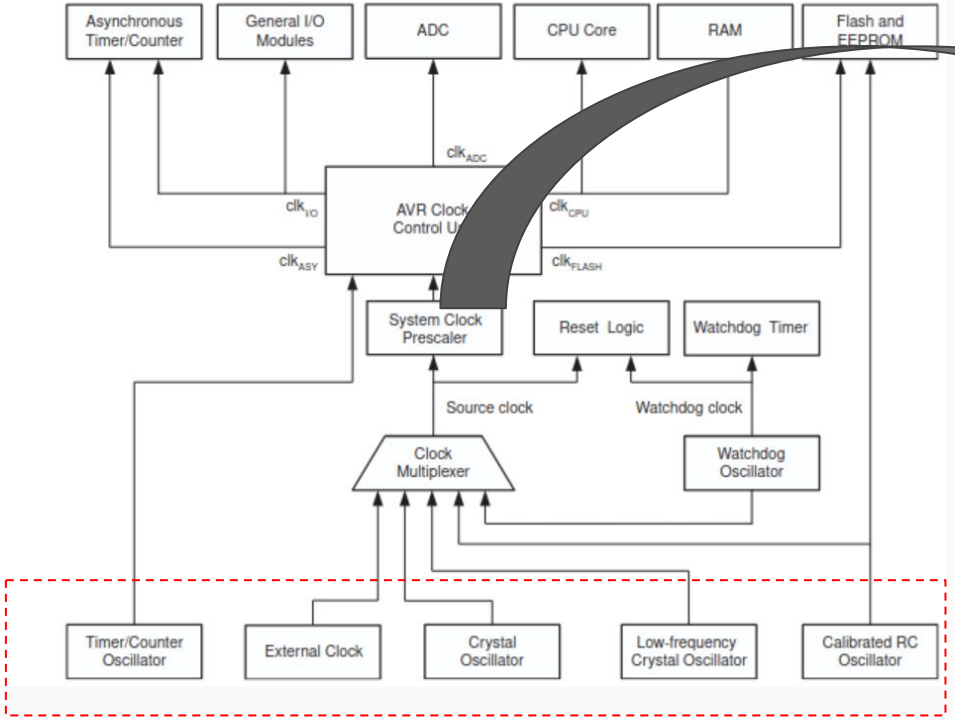
Fontes de clock

externo (cristal), interno,...

A fonte deve ser selecionado nos registradores específicos

ATMega328 - Sistema de clock

Figure 9-1. Clock Distribution

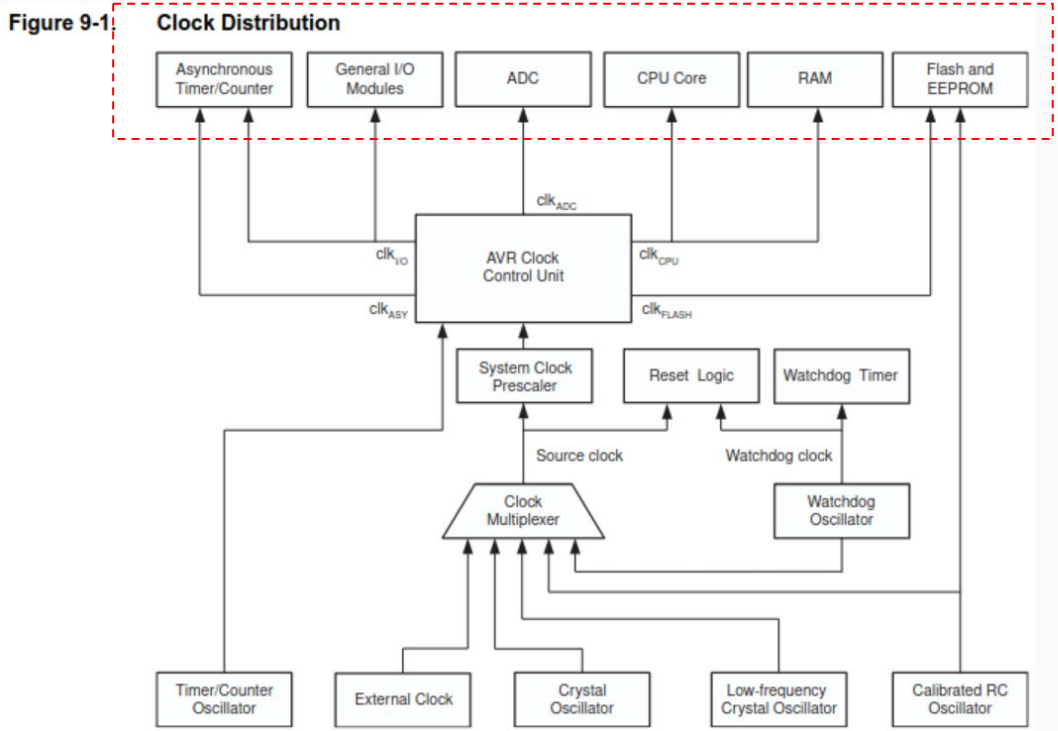


Prescaler

Divisão de frequência

Possibilita trabalhar com frequências de clock internas diferentes do clock externo

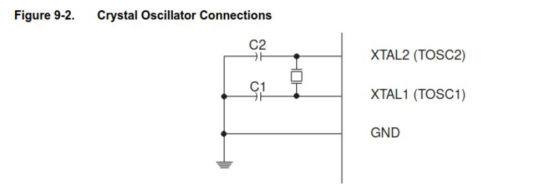
ATMega328 - Sistema de clock



Blocos de periféricos

- Podem trabalhar com frequência de clock <u>diferentes da frequência de clock da CPU</u>
- Otimização do consumo de energia - periféricos mais lentos (menor consumo de energia) ou mais rápidos (maior consumo)

Conexões elétricas:



Atividade preparatória - Avaliação teórica MIC

Lista de exercício 03 - moodle

8 questões
Arquitetura do AVR

Individual

Prazo: 15 dias (30/03)

Materiais de referência:

Vídeo-aulas

- Características microcontroladores AVR
- Arquitetura Interna do AVR
- Memórias internas
- Sistemas de clock e reset

Data sheet - ATmega328

MICROCHIP

ATmega48A/PA/88A/PA/168A/PA/328/P

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Livro

Livro - AVR - Técnicas de projeto